# Hodge Structure and Gauge Equivalence of $\ell^1$ Defect Fields

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## Classification of Results

This paper separates rigorous mathematics from original constructions and speculative analogies. Each result is annotated with its epistemic layer:

| Layer | Status | What it contains |
|---|---|---|
| **A: Theorems** | Established mathematics | Standard results from discrete algebraic topology and discrete $\ell^2$ Hodge theory on finite graphs without 2-cells [1, 7, 8]. |
| **B: Construction** | Original to this paper | The Cohomological $\ell^1$ Norm $\Phi$ (terminology adopted here for the $\ell^1$ quotient norm on graph cohomology), the proof of its sparsity bias, LP cut-duality, and the inflation quantification. |
| **C: Analogy** | Speculative | The structural parallel to anisotropic total variation (graph-TV) denoising in image processing [2, 13]. |

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## Abstract

The preceding works [14, 15, 16] establish that additive, structure-preserving diagnostics on coboundary spaces uniquely induce an $\ell^1$ geometry under explicit axioms. Given this constraint, a natural question is how to define a gauge-invariant measure of the irreducible component of a defect field.

On finite graphs, classical discrete Hodge theory provides an $\ell^2$-orthogonal decomposition separating exact and harmonic components. However, this decomposition is not compatible with the $\ell^1$ metric: the $\ell^2$-orthogonal projection onto the harmonic space does not, in general, minimize $\ell^1$ cost within a cohomology class.

We study the functional

$$\Phi([\delta]) = \min_{f \in C^0} \|\delta - d_0 f\|_1,$$

which is the standard $\ell^1$ quotient norm on $C^1/\operatorname{im}(d_0)$ (the minimum is attained; see Proposition 8.2). While this optimization problem is classical (appearing in

total variation minimization and minimum homologous chain problems [12]), we show that, under the structural constraints identified in [14–16], it is the unique gauge-invariant norm compatible with the forced $\ell^1$ geometry, in the sense that any norm on $H^1(G)$ induced from the $\ell^1$ norm on $C^1$ via the quotient map must equal $\Phi$ (Theorem 8.1, Remark 8.6).

We further relate $\Phi$ to discrete Hodge decomposition, prove that $\ell^2$ projection generically inflates $\ell^1$ cost (with an explicit ratio of $3 - 2/n$ on cycle graphs $C_n$), and characterize $\Phi$ via linear programming duality as the dual norm of bounded divergence-free circulations. This provides a unified interpretation connecting cohomology, $\ell^1$ optimization, and network flows, without introducing new optimization machinery.

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## 1. Introduction

The preceding papers in this series isolated the exact geometric magnitude of interaction defects generated when correlated systems are projected onto locally factorized manifolds. Paper 002 [16] established that coordinate-wise additivity forces an $\ell^1$ geometry on finite graph cochains. Paper 001 [15] generalized this to Banach presheaves via functoriality under bounded linear maps. Paper 000 [14] applied this geometric rigidity to show that projecting dynamics onto a product-state manifold creates an $\ell^1$-measurable curvature obstruction that persists at finite density.

A natural next question arises: given a defect field $\delta \in C^1(G)$, how should one decompose it into removable (exact) and irremovable (cohomological) components? The classical tool is the discrete Hodge decomposition, which operates in $\ell^2$. But since the preceding papers establish $\ell^1$ as the uniquely forced diagnostic geometry, a tension emerges.

Crucially, the preceding papers do not merely motivate the use of an $\ell^1$ metric — they *force* it. Throughout, we assume the $\ell^1$ geometry established in [14–16]; this paper studies its consequences. Any decomposition or invariant associated with defect fields must therefore be compatible with this geometry. The central problem addressed here is thus not to choose a convenient norm, but to determine the canonical cohomological structure induced by the uniquely determined $\ell^1$ metric.

**Classical Hodge decomposition does not generally hold in $L^1$ Banach spaces**, as they lack the inner-product structure necessary to define orthogonal projections. We do not construct an $\ell^1$ Hodge decomposition. Instead, we use the classical $\ell^2$ Hodge decomposition as an algebraic coordinate system on $C^1(G)$ and evaluate its components under the $\ell^1$ norm. By operating on **finite graphs** (finite-dimensional discrete 1-complexes), the vector space topology ensures that the $\ell^2$ decomposition remains algebraically available, while the resulting mismatch between orthogonal projection in $\ell^2$ and sparsity structure

in $\ell^1$ is made precise.

We demonstrate that while the discrete $\ell^2$ Hodge operators dictate the irremovable topological *shape* of the obstruction, they generically increase its $\ell^1$ *magnitude*. To capture the true physical penalty, we construct the **Cohomological $\ell^1$ Norm**: a well-defined functional $\Phi : H^1(G, \mathbb{R}) \to \mathbb{R}_{\geq 0}$ that evaluates the topological defect not by $\ell^2$ projection, but by infimal $\ell^1$ gauge cost.

**Scope and Relation to Existing Work.** We emphasize that the optimization problem $\min_f \|\delta - d_0 f\|_1$ is not new. It appears in several established contexts, including anisotropic total variation on graphs [2, 13], $\ell^1$ regression over incidence matrices [4], the minimum homologous chain problem [12], and is related to Gromov's bounded cohomology [11] in the infinite-dimensional setting. Compressed sensing and basis pursuit [5] exploit the same $\ell^1$ sparsity structure, and shortest cycle basis algorithms [6] address the underlying combinatorial structure. The contribution of this paper is not to introduce a new optimization framework, but to identify this functional as the unique gauge-invariant norm compatible with the $\ell^1$ geometry that is structurally forced under the assumptions of [14–16]. Accordingly, the results here should be understood as a structural interpretation and organization of known optimization objects within a cohomological framework, rather than as the introduction of new algorithms or complexity results.

**Recall (Papers 001–002).** The $\ell^1$ coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is the unique (up to positive scalar) edge-additive, faithful, functorial seminorm on the 1-cochain space $C^1(P, F)$ [15, 16]. In the presheaf language of [15], the coboundary is $\delta_e(s) = r_{b,a}(s(b)) - s(a)$. When the Banach presheaf $F$ is the constant sheaf $\mathbb{R}$ on a graph $G$, this reduces to $(d_0 f)(e) = f(v) - f(u)$, the total norm $I(s)$ becomes $\|d_0 f\|_1$, and the quotient norm $\Phi([\delta]) = \min_f I(s - d_0 f)$ measures the irreducible cost after gauge optimization. This paper develops the structure of $\Phi$.

**Notation bridge.** The following dictionary connects the presheaf notation of [14, 15] to the graph notation used in the remainder of this paper:

| Presheaf notation (Papers 000–001) | Graph notation (this paper) |
|---|---|
| Poset $P$, covering relations $E_{\mathrm{cov}}(P)$ | Graph $G = (V, E)$ |
| Banach presheaf $F$, stalks $F(x)$, restrictions $r_{b,a}$ | Constant sheaf $\mathbb{R}$, identity restrictions |
| Coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ | Discrete derivative $(d_0 f)(e) = f(v) - f(u)$ |
| $\ell^1$ coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ | $\|d_0 f\|_1 = \sum_e \lvert(d_0 f)_e\rvert$ |

**Conventions.** Throughout this paper, $G = (V, E)$ denotes a finite, connected, undirected graph. Each edge is assigned an arbitrary but fixed orientation, used only for algebraic bookkeeping; all results are independent of this choice. We work

over the scalar field $\mathbb{R}$. Norms are denoted $\|\cdot\|$ with appropriate subscripts ($\ell^1$, $\ell^2$, $\ell^\infty$); absolute values of scalars are denoted $|\cdot|$. The extension to vector-valued defects $\delta \in C^1(G, \mathbb{R}^k)$ proceeds component-wise and is discussed in Section 10.

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## 2. Gauge Equivalence (Layer A/B)

Before constructing the decomposition machinery, we introduce the central physical notion that motivates the entire analysis.

A defect field $\delta \in C^1(G)$ records the local projection error across each edge. However, the absolute value of $\delta$ on any single edge is not physically meaningful: it depends on the choice of local reference state at each vertex. A **gauge transformation** is a reassignment of these local reference states, parameterized by a vertex function $f \in C^0(G) = \mathbb{R}^{|V|}$. We use "gauge equivalence" in the sense of equivalence under reassignment of local reference frames, analogous to the gauge freedom of electromagnetic potentials $A \mapsto A + d\Lambda$.

**Definition 2.1 (Gauge Equivalence).** Two defect fields $\delta, \delta' \in C^1(G)$ are **gauge equivalent** if their difference is an exact gradient:

$$\delta \sim \delta' \iff \delta - \delta' \in \operatorname{im}(d_0)$$

where $d_0 : C^0 \to C^1$ is the discrete exterior derivative defined in Section 3. The equivalence class $[\delta] = \delta + \operatorname{im}(d_0)$ is the **gauge class** of $\delta$.

The physically meaningful quantity is not $\delta$ itself but its gauge class $[\delta]$. The natural gauge-invariant magnitude of an obstruction is therefore:

$$\operatorname{Cost}(\delta) = \min_{f \in C^0} \|\delta - d_0 f\|_1$$

This is precisely the $\ell^1$ quotient norm on $C^1/\operatorname{im}(d_0)$. It represents the minimal $\ell^1$ cost achievable over all gauge-equivalent representatives. We will show that this quantity defines a norm on the first cohomology group $H^1(G, \mathbb{R})$ and admits a clean dual formulation connecting it to the topological cycle structure of $G$.

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## 3. The Discrete de Rham Complex (Layer A)

Let $C^0 = \mathbb{R}^{|V|}$ denote the space of 0**-cochains** (vertex functions) and $C^1 = \mathbb{R}^{|E|}$ denote the space of 1**-cochains** (edge fields). Each edge $e$ carries its fixed orientation $e = (u, v)$.

**Definition 3.1 (Discrete Exterior Derivative).** The operator $d_0 : C^0 \to C^1$ is the graph difference operator:

$$(d_0 f)(e_{uv}) = f(v) - f(u)$$

Its matrix representation is the signed incidence matrix $B \in \mathbb{R}^{|E|\times|V|}$, with $B_{e,v} = +1$, $B_{e,u} = -1$ for each oriented edge $e = (u, v)$.

**Definition 3.2 (Discrete Codifferential / Divergence).** The adjoint $d_0^* : C^1 \to C^0$ is defined with respect to the standard $\ell^2$ inner products on $C^0$ and $C^1$:

$$\langle d_0 f, \omega\rangle_{\ell^2} = \langle f, d_0^*\omega\rangle_{\ell^2}$$

Explicitly, $d_0^* = B^T$ and the **divergence** of a 1-cochain $\omega$ at a vertex $u$ is the net signed flow:

$$(d_0^*\omega)(u) = \sum_{e \text{ incoming at } u} \omega(e) - \sum_{e \text{ outgoing from } u} \omega(e)$$

The discrete de Rham complex for a graph without 2-cells is:

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{0} 0$$

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## 4. The Hodge Laplacian and Harmonic Cochains (Layer A)

Following Eckmann [1], Horak–Jost [7], and Lim [8], the full 1-form Laplacian on a simplicial complex is:

$$\Delta_1 = d_0\, d_0^* + d_1^*\, d_1$$

**Remark 4.1 (Convention and Missing 2-Cells).** For a graph (a 1-dimensional simplicial complex), there are no 2-cells. Consequently $d_1 \equiv 0$, and the Laplacian collapses to:

$$\Delta_1 = d_0\, d_0^* \quad \text{(graphs without 2-cells)}$$

**Definition 4.2.** A 1-cochain $\omega \in C^1$ is **harmonic** if $\Delta_1\omega = 0$: $\mathcal{H}^1(G) = \ker \Delta_1$.

**Proposition 4.3.** *For a graph without* 2*-cells,* $\ker \Delta_1 = \ker d_0^*$.

*Proof.* $\ker d_0^* \subseteq \ker \Delta_1$ is immediate. Conversely, $\Delta_1\omega = 0$ implies $0 = \langle d_0 d_0^*\omega, \omega\rangle_{\ell^2} = \|d_0^*\omega\|_{\ell^2}^2$, so $d_0^*\omega = 0$. ■

**Proposition 4.4.** *If $G$ is connected,* $\dim \mathcal{H}^1(G) = \beta_1(G) = |E| - |V| + 1$.

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## 5. The Discrete Hodge Decomposition (Layer A)

On a general continuous manifold, the Hodge theorem splits any 1-form into three orthogonal components: $\omega = d\alpha + \delta\beta + h$ (exact, coexact, and harmonic). However, because our state space is a finite graph with no 2-cells, $d_1 = 0$, its adjoint $d_1^* = 0$, and the coexact term vanishes entirely. The decomposition reduces to exactly two functional pieces.

**Theorem 5.1 (Bipartite Hodge Decomposition).** *Every $\delta \in C^1(G)$ decomposes algebraically uniquely as:*

$$\delta = \delta_{\text{ex}} + \delta_{\text{harm}}, \quad \delta_{\text{ex}} \in \text{im}(d_0), \quad \delta_{\text{harm}} \in \ker d_0^*, \quad \delta_{\text{ex}} \perp_{\ell^2} \delta_{\text{harm}}$$

*Proof.* By the orthogonal complement theorem in finite dimensions, $C^1 = \text{im}(d_0) \oplus (\text{im}(d_0))^\perp$. Since $\langle d_0 f, h\rangle_{\ell^2} = \langle f, d_0^* h\rangle_{\ell^2}$ for all $f, h$, we have $(\text{im}(d_0))^\perp = \ker(d_0^*)$. Therefore $C^1 = \text{im}(d_0) \oplus \ker(d_0^*)$ with the two summands $\ell^2$-orthogonal. ■

**Remark 5.2 (Finite-Dimensional Availability).** This theorem explicitly relies on the finite-dimensionality of the graph. Infinite-dimensional $L^1$ spaces lack orthogonal complements, making $L^1$ Hodge theory notoriously pathological. We do not attempt an $\ell^1$ Hodge decomposition. Instead, we use the $\ell^2$ Hodge decomposition as an algebraic coordinate system on $C^1(G)$ — guaranteed to exist in finite dimensions — and then evaluate its components under the $\ell^1$ norm. The resulting mismatch reflects the incompatibility between orthogonal projection in $\ell^2$ and sparsity structure in $\ell^1$.

**Remark 5.3 (Topological Interpretation).** Under this framework, $\text{im}(d_0)$ corresponds to potential difference relationships (gauge transformations), while $\ker d_0^*$ captures network circulations. The harmonic component $\delta_{\text{harm}}$ is the gauge-invariant core, unchanged under $\delta \mapsto \delta - d_0 g$. For trees ($\beta_1 = 0$), all defects are exact and trivially gauge-removable. For graphs with loops ($\beta_1 > 0$), any non-zero harmonic component constitutes an irreducible topological obstruction.

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## 6. The $\ell^1/\ell^2$ Tension (Layer A/B)

**Remark 6.1 (Layer A).** $\|x + y\|_1 \leq \|x\|_1 + \|y\|_1$, with equality iff $x_e y_e \geq 0$ for all $e$. This is the standard triangle inequality.

**Remark 6.2 (Layer A).** Applying the triangle inequality to the Hodge decomposition: $\|\delta\|_1 \leq \|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1$, with equality iff there is edgewise sign compatibility between the exact and harmonic components.

**Remark 6.3.** Generic sign conflicts make the inequality strict. Thus, the $\ell^2$ algebraic Hodge projection operator generically yields strictly larger $\ell^1$ component sums except in sign-compatible cases.

**Proposition 6.4 (Failure of $\ell^2$ Hodge Minimization for $\ell^1$ Cost) (Layer B).** Let $\delta \in C^1(G)$ and let $\delta = \delta_{\text{ex}} + \delta_{\text{harm}}$ be its $\ell^2$ Hodge decomposition. Then:

1. The harmonic component does not, in general, minimize $\ell^1$ cost over the cohomology class $[\delta]$:
$$\|\delta_{\text{harm}}\|_1 \geq \Phi([\delta]),$$
with strict inequality for generic $\delta$.

2. Equality holds if and only if $\delta_{\text{harm}}$ is already an $\ell^1$-optimal representative of its cohomology class, which requires sign compatibility between the harmonic component and any exact correction.

In particular, the $\ell^2$-orthogonal projection onto $\ker d_0^*$ is not an $\ell^1$-optimal representative of the cohomology class.

*Proof.* By definition, $\Phi([\delta]) = \min_{f \in C^0} \|\delta - d_0 f\|_1 \leq \|\delta_{\text{harm}}\|_1$ since $\delta_{\text{harm}} = \delta - d_0 f_{\ell^2}$ for some $f_{\ell^2}$. Strict inequality holds whenever a different gauge representative achieves lower $\ell^1$ cost. By the crosspolytope geometry of the $\ell^1$ ball, the $\ell^2$-optimal representative (which minimizes Euclidean distance to $\ker d_0^*$) generically diffuses mass across coordinates, while the $\ell^1$-optimal representative concentrates mass on minimal support. These coincide only when the harmonic projection is already sparse (sign-compatible). The examples below demonstrate strict inequality on all cycle graphs $C_n$, $n \geq 3$. ■

### 6.1 Triangle Graph (Layer B)

**Example 6.5.** On $C_3$ with $\beta_1 = 1$, cycle vector $c = (1,1,1)^T$. A sparse local defect $\delta = (1,0,0)^T$ resolves via $\ell^2$ projection to: - $\delta_{\text{harm}} = \frac{1}{3}(1,1,1)^T$, $\delta_{\text{ex}} = \frac{1}{3}(2,-1,-1)^T$ - $\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1 = \frac{4}{3} + 1 = \frac{7}{3} > 1 = \|\delta\|_1$ - Inflation ratio: $7/3 \approx 2.33$

### 6.2 Square Cycle (Layer B)

**Example 6.6.** On $C_4$ with $\delta = (1,0,0,0)^T$: - $\delta_{\text{harm}} = \frac{1}{4}(1,1,1,1)^T$, $\delta_{\text{ex}} = \frac{1}{4}(3,-1,-1,-1)^T$ - $\|\delta_{\text{harm}}\|_1 = 1$, $\|\delta_{\text{ex}}\|_1 = \frac{3}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{6}{4} = \frac{3}{2}$ - $\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1 = \frac{5}{2} > 1 = \|\delta\|_1$ - Inflation ratio: $5/2 = 2.5$

### 6.3 Inflation Ratio on Cycle Graphs (Layer B)

**Proposition 6.7 ($\ell^2$ Inflation Ratio on $C_n$).** *For a single-edge defect $\delta = \mathbf{e}_1 = (1,0,\ldots,0)^T$ on the cycle graph $C_n$ ($n \geq 3$), the $\ell^2$ Hodge decomposition inflation ratio is:*

$$\frac{\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1}{\|\delta\|_1} = 3 - \frac{2}{n}$$

*Proof.* The harmonic space of $C_n$ is $\ker(d_0^*) = \operatorname{span}(\mathbf{1})$, where $\mathbf{1} = (1, 1, \ldots, 1)^T$. The $\ell^2$-orthogonal projection onto $\ker(d_0^*)$ gives:

$$\delta_{\text{harm}} = \frac{\langle \delta, \mathbf{1} \rangle}{\langle \mathbf{1}, \mathbf{1} \rangle} \mathbf{1} = \frac{1}{n} \mathbf{1}$$

$$\delta_{\text{ex}} = \delta - \delta_{\text{harm}} = \left( \frac{n-1}{n}, \frac{-1}{n}, \ldots, \frac{-1}{n} \right)$$

Computing the $\ell^1$ norms:

$$\|\delta_{\text{harm}}\|_1 = n \cdot \frac{1}{n} = 1$$

$$\|\delta_{\text{ex}}\|_1 = \frac{n-1}{n} + (n-1) \cdot \frac{1}{n} = \frac{2(n-1)}{n}$$

The inflation ratio is:

$$\frac{\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1}{\|\delta\|_1} = \frac{\frac{2(n-1)}{n} + 1}{1} = \frac{2n - 2 + n}{n} = \frac{3n-2}{n} = 3 - \frac{2}{n}$$

This grows monotonically with $n$ and approaches 3 as $n \to \infty$. ■

**Remark 6.8 (Crosspolytope Bias).** There is a geometric reason why the $\ell^2$ harmonic projection generically increases $\ell^1$ magnitude. The mismatch stems from the fundamental difference between Euclidean balls ($\ell^2$) and crosspolytopes ($\ell^1$) in high dimensions. The $\ell^2$ orthogonal projection minimizes Euclidean distance but naturally diffuses mass across coordinates uniformly (e.g., $1/3, 1/3, 1/3$). The $\ell^1$ cost, dictated by the crosspolytope geometry, privileges sparse, extreme configurations (corners). When mapping a sparse exact defect onto the harmonic cycle space, $\ell^2$ projection destroys its sparsity – an inflation directly proportional to the loss of corner-localization.

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## 7. The Cohomological $\ell^1$ Norm (Layer B)

This section develops the **Cohomological $\ell^1$ Norm**, a term we adopt for the $\ell^1$ quotient norm on $C^1/\operatorname{im}(d_0)$. The quotient norm construction is standard in functional analysis [4, 9]; the $\ell^1$ quotient norm on homology classes appears in the bounded cohomology literature [11] and the minimum homologous chain problem [12]. The contribution here is not the optimization framework itself, but the identification, under the additivity and functoriality assumptions established in [14–16], of $\Phi$ as the unique gauge-invariant norm on $H^1(G)$ induced by the forced $\ell^1$ geometry (Remark 8.6).

### 7.1 The $\ell^1$ Quotient Norm

As established in Section 2, the scalar magnitude invariant under arbitrary local reference rescalings is the $\ell^1$ **quotient norm** on the space of cochains modulo exact gradients $C^1/\operatorname{im}(d_0)$:

$$\mathrm{Cost}(\delta) = \min_{f \in C^0} \|\delta - d_0 f\|_1$$

### 7.2 Descent to Cohomology

The gauge class $[\delta] = \delta + \operatorname{im}(d_0)$ contains a unique element of $\ker d_0^*$, namely $\delta_{\mathrm{harm}} = \Pi_{\ker d_0^*}(\delta)$, the $\ell^2$-orthogonal projection. This identification uses the $\ell^2$ Hodge decomposition to select a canonical representative within each cohomology class; the $\ell^1$ minimization itself is independent of this choice. Therefore:

$$\mathrm{Cost}(\delta) = \min_{r \in \delta_{\mathrm{harm}} + \operatorname{im}(d_0)} \|r\|_1$$

This evaluates to a purely cohomological constraint. It depends only on the cohomology class $[\delta]$, with the harmonic representative providing a canonical coordinate in $H^1(G, \mathbb{R})$.

### 7.3 The Cohomological Norm

**Proposition 7.1 (Existence of Minimizers).** *The minimum* $\min_{f \in C^0} \|\delta - d_0 f\|_1$ *is attained for every* $\delta \in C^1(G)$.

*Proof.* The objective $f \mapsto \|\delta - d_0 f\|_1$ is continuous and coercive on $C^0/\ker(d_0) \cong \mathbb{R}^{|V|-1}$ (since $\ker(d_0) = \operatorname{span}(\mathbf{1})$ for connected $G$ and $\|d_0 f\|_1 \to \infty$ as $\|f\|_2 \to \infty$ on this quotient). A continuous coercive function on a finite-dimensional space attains its infimum [9]. ■

*Every minimizer* $r^* = \delta - d_0 f^*$ *satisfies the first-order optimality condition* $B^T s^* = 0$ *where* $s^* \in \partial \|\cdot\|_1(r^*)$ *is a subgradient [9, Section 23]. That is,* $s^*$ *is a divergence-free flow bounded entrywise by 1.*

**Theorem 7.2 (Cohomological $\ell^1$ Norm).** *The functional*

$$\Phi : H^1(G, \mathbb{R}) \to \mathbb{R}_{\geq 0}, \qquad \Phi([\delta]) = \min_{r \in [\delta]} \|r\|_1$$

*is a norm on* $H^1(G, \mathbb{R})$.

*Proof.* **Positive definiteness.** If $\Phi([\delta]) = 0$, then by Proposition 7.1 the minimum is attained: there exists $f^*$ with $\|\delta - d_0 f^*\|_1 = 0$, hence $\delta = d_0 f^*$, so $[\delta] = 0$. Conversely, $[\delta] = 0$ gives $\delta \in \operatorname{im}(d_0)$, so $\Phi([\delta]) = 0$.

**Absolute homogeneity.** For $\lambda \neq 0$: $\Phi([\lambda\delta]) = \min_f \|\lambda\delta - d_0 f\|_1 = \min_f \|\lambda(\delta - d_0(f/\lambda))\|_1 = |\lambda| \min_g \|\delta - d_0 g\|_1 = |\lambda| \Phi([\delta])$, where $g = f/\lambda$. For $\lambda = 0$: $\Phi([0]) = 0$.

**Triangle inequality.** Let $f^*, g^*$ be minimizers for $[\delta]$ and $[\delta']$ respectively (existing by Proposition 7.1). Choosing $f = f^* + g^*$ in the minimization for $[\delta + \delta']$:

$$\Phi([\delta+\delta']) = \min_f \|\delta+\delta'-d_0 f\|_1 \leq \|(\delta-d_0 f^*)+(\delta'-d_0 g^*)\|_1 \leq \|\delta-d_0 f^*\|_1+\|\delta'-d_0 g^*\|_1 = \Phi([\delta])+\Phi([\delta'])$$

■

### 7.4 Sparsity Bias

**Remark 7.3 (Geometric Sparsity Bias).** $\ell^1$ minimization inherently contacts the bounding faces of the crosspolytope geometry on low-dimensional faces (minimizing support). Consequently, optimization concentrates defects onto minimal edge cycles rather than expanding them as uniform fractional obstructions like the $\ell^2$ harmonic projection inevitably does.

### 7.5 Structure on Cycle Bases and Minimum Weight Combinations

While minimizing $\ell^1$ cost generally necessitates linear programming, the structural geometry of the cycle space allows for a direct evaluation on the graph's loops under favorable conditions.

**Theorem 7.4 (Evaluation on Edge-Disjoint Cycles).** *Let $\{C_1, \ldots, C_{\beta_1}\}$ be a cycle basis for $H^1(G)$ in which the cycles are pairwise edge-disjoint and vertex-disjoint (i.e., no two cycles share an edge or a vertex). Given the harmonic expansion $\delta_{harm} = \sum_{i=1}^{\beta_1} a_i C_i$, the cohomological norm is:*

$$\Phi([\delta]) = \sum_{i=1}^{\beta_1} |a_i| \|C_i\|_1$$

*Proof.* Since the cycles are vertex-disjoint, their vertex sets $V_i$ are pairwise disjoint. The potential $f \in C^0$ decomposes as $f = \sum_i f_i$ where $f_i$ is supported on $V_i$. The exact form $d_0 f$ restricted to edge set $E_i$ depends only on $f_i$ (vertices outside $V_i$ contribute zero to edges in $E_i$). Therefore:

$$\min_f \|\delta - d_0 f\|_1 = \sum_{i=1}^{\beta_1} \min_{f_i \in \mathbb{R}^{V_i}} \sum_{e \in E_i} |a_i (C_i)_e - (d_0 f_i)_e|$$

On each component, the cohomology class $[a_i C_i]$ is one-dimensional, and any exact correction on a single cycle changes the representative by $d_0 f_i$ for $f_i$ supported on $V_i$. Since $C_i$ is a single cycle, the minimum $\ell^1$ representative of the class $[a_i C_i]$ has cost $|a_i| \|C_i\|_1$ (any exact correction on a cycle merely redistributes, never reduces, the total absolute circulation). ■

**Remark 7.5.** When cycles share vertices but not edges, the decomposition may still hold in practice, but the proof requires a more delicate LP analysis

showing that the shared-vertex coupling constraints are independently satisfiable. When cycles share edges, the general case requires the linear program to resolve interactions, corresponding to the **minimum weight homologous chain** [12].

**Remark 7.6 (Geometry of the Cycle Polytope).** Under the $\Phi$ functional, the metric geometry of $H^1(G)$ behaves structurally as a polyhedral metric space whose unit locus mirrors the network's bounded cycle polytope $(\mathrm{conv}\{\pm C_1, \ldots, \pm C_{\beta_1}\})$.

**Remark 7.7 (Uniqueness of $\Phi$).** The norm $\Phi$ is the unique norm on $H^1(G)$ induced via the $\ell^1$ quotient construction on $C^1(G)$ through the quotient map $C^1 \to C^1/\operatorname{im}(d_0)$. This follows from the standard construction: the quotient norm $\|[x]\| = \inf\{\|y\| : y \in [x]\}$ is the largest norm on the quotient space making the projection non-expansive [9, Section 2]. Since [14–16] establish $\ell^1$ as the uniquely forced norm on $C^1$, the quotient construction uniquely determines $\Phi$. Choosing any other gauge-invariant norm on $H^1$ (e.g., the $\ell^2$ norm on harmonic representatives) would break compatibility with the induced $\ell^1$ geometry.

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## 8. Total Variation Denoising and Fast Algorithms (Layer C)

The gauge optimization $\min_f \|\delta - Bf\|_1$ exhibits structural isomorphism to **anisotropic total variation (TV) denoising** extensively developed in image filtering [2, 13]. Although the optimization is formally identical to graph total variation minimization, its role here is different: it arises as the quotient norm induced by the uniquely determined $\ell^1$ geometry on coboundary spaces [14–16], rather than as a regularization principle or signal recovery objective. Setting the mapping between computational models:

| Imaging | Defect Topology |
|---|---|
| Image Signal | Vertex Potential ($f$) |
| Image Gradient ($\nabla$) | Exterior Derivative ($d_0 = B$) |
| Observed Gradient ($y$) | Source Defect Field ($\delta$) |
| TV Residual ($y - \nabla u$) | Topological Obstruction ($r$) |

Consequently, the optimization objective directly mirrors **graph total variation (graph-TV)**. Where classical TV identifies structural borders (sparse semantic image edges), the framework localizes purely topological features (**sparse cycle contradictions**).

**Computational Remark.** This overlap permits the use of algorithms designed for large-scale imaging tasks. Rather than utilizing generic $O(|E|^3)$ interior point bounds, the $\ell^1$ quotient norm is computable using splitting protocols

(Chambolle–Pock primal-dual [2], Split Bregman, ADMM), which admit per-iteration complexity scaling linearly in $|E|$. Whether these methods achieve comparable convergence rates in the graph cohomology setting is an empirical question not addressed here; we note only the formal structural correspondence.

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## 9. Cut Duality and Flow Circulation (Layer B)

We establish that the topological magnitude of the norm is strictly characterized by valid internal network configurations.

**Theorem 9.1 (Cut Duality).**

$$\Phi([\delta]) = \max_{\phi} \{\langle \phi, \delta \rangle : \|\phi\|_\infty \leq 1,\ d_0^* \phi = 0\}$$

*Proof.* The primal problem $\min_f \|\delta - Bf\|_1$ is reformulated as a linear program by introducing auxiliary variables $t_e \geq 0$:

$$\text{Primal: } \min_{f,t} \sum_e t_e \quad \text{subject to} \quad -t_e \leq \delta_e - (Bf)_e \leq t_e, \quad t_e \geq 0$$

Let $\phi_e^+, \phi_e^- \geq 0$ be the dual variables for the upper and lower bound constraints respectively. Setting $\phi_e = \phi_e^+ - \phi_e^-$, the dual problem becomes:

$$\text{Dual: } \max_{\phi} \langle \phi, \delta \rangle \quad \text{subject to} \quad B^T \phi = 0, \quad |\phi_e| \leq 1 \text{ for all } e$$

The constraint $B^T \phi = d_0^* \phi = 0$ requires $\phi$ to be divergence-free (a circulation). The bound $|\phi_e| \leq 1$ constrains $\phi \in B_\infty$. Strong duality holds because the problem is a finite-dimensional LP with a non-empty primal feasible set ($f = 0$ is feasible) and a bounded primal objective [4, Chapter 5]. ■

**Remark 9.2 (Dual Norm of Bounded Circulations).** The $d_0^* \phi = 0$ condition fixes $\phi \in Z_1(G)$, the cycle space. The bounded infinity constraint yields:

$$\Phi([\delta]) = \max_{\phi \in Z_1(G) \cap B_\infty} \langle \phi, \delta \rangle$$

Hence, $\Phi$ resolves as the exact **dual norm of bounded circulations**. Within the network interpretation, it isolates the maximal potential flow pairing compatible with the measured contradiction field $\delta$.

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## 10. Extension to Vector-Valued Defects (Layer A/B)

For $\delta \in C^1(G, \mathbb{R}^k)$, define the separable $\ell^1$ norm $\|\delta\|_1 = \sum_{e \in E} \sum_{j=1}^{k} |\delta_{e,j}|$ (i.e., $\ell^1$ over edges, then $\ell^1$ over components). Under this norm, the decomposition and cohomological norm extend component-wise: $\Phi([\delta]) = \sum_{j=1}^{k} \Phi([\delta^{(j)}])$.

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## 11. Discussion

### 11.1 What This Paper Establishes

The discrete $\ell^2$ Hodge decomposition canonically identifies the topological shape of a cohomology class, but it does not respect the $\ell^1$ geometry established in the preceding papers [14, 15, 16] as the uniquely forced diagnostic norm. The Cohomological $\ell^1$ Norm $\Phi([\delta])$ resolves this by optimizing over gauge-equivalent representatives to find the one of minimal $\ell^1$ cost.

The conceptual progression is: 1. **Topological classification** (Theorem 5.1, Layer A): Finite dimensions rescue the algebraic Hodge decomposition, categorizing gauge-removable and irreducible obstructions. 2. $\ell^1/\ell^2$ **tension** (Section 6, Layer B): Evaluates Euclidean vs crosspolytope geometry, quantifying the harmonic $\ell^2$ projection cost inflation at a ratio of $3 - 2/n$ on cycle graphs. 3. **Cohomological norm** (Theorem 7.2, Layer B): Formalizes $\Phi([\delta]) = \min_r \|r\|_1$ as the $\ell^1$ quotient norm evaluating the defect's cycle space structure. 4. **Circulation duality** (Theorem 9.1, Layer B): Expresses obstruction severity as the maximal bounded divergence-free flow pairing. 5. **Image processing parallelism** (Section 8, Layer C): Notes the structural correspondence to TV denoising methods.

### 11.2 Relation to Prior Work

The discrete de Rham complex on graphs is due to Eckmann [1]. The bipartite Hodge decomposition on graphs without 2-cells is standard (see Lim [8] for a modern treatment). The LP duality formulation of $\Phi$ connects to classical network flow theory (Ford–Fulkerson [3]) and the coarea formula for graph total variation (Chambolle et al. [2]). The $\ell^1$ quotient norm on cohomology is related to Gromov's bounded cohomology [11], though the finite-dimensional setting here avoids the subtleties of the infinite-dimensional theory. The quotient norm construction itself is standard in functional analysis [9]. The discrete calculus framework is developed systematically in [10].

### 11.3 Position in the Unified Program

This paper completes the structural program initiated in the preceding works by resolving the final open question: given that $\ell^1$ is the uniquely forced geometry on coboundary defects, what is the canonical gauge-invariant decomposition and measurement of such defects?

The progression is now complete:

1. **Combinatorial Level (Paper 002)** [16]: Local additivity over disjoint coordinate supports and edge separability force $\ell^1$ geometry on finite cochains.
2. **Functional Level (Paper 001)** [15]: Structural compatibility (functoriality under bounded linear maps) extends this rigidity to Banach presheaves.
3. **Dynamical Level (Paper 000)** [14]: Projection of linear dynamics onto nonlinear manifolds generates intrinsic coboundary defects measured in this geometry, persisting at non-vanishing density.
4. **Cohomological Level (this paper)**: The forced $\ell^1$ geometry induces a canonical quotient norm $\Phi$ on $H^1$, providing the unique gauge-invariant measure of irreducible obstruction (Remark 7.7). The $\ell^2$ Hodge decomposition identifies the topological shape; $\Phi$ measures the true $\ell^1$ cost.

Thus, the $\ell^1$ structure is not an isolated phenomenon but persists across combinatorial, functional, dynamical, and topological layers. The measurement, decomposition, and optimization of defect fields are not independent choices, but are constrained by a single underlying geometric requirement.

*Remark (Observer-theoretic grounding).* The axioms forcing $\ell^1$ geometry at each level of this escalation ladder admit an interpretation in terms of observational consistency: they correspond to the minimal structural conditions under which a local agent can consistently detect and aggregate defects. Edge locality requires local evaluability; coordinate additivity requires that independent channels contribute without cancellation; symmetry requires indifference to relabeling of structurally identical subsystems. Under this reading, the four-level program — from combinatorial cochains through Banach presheaves and dynamics to the cohomological quotient norm developed here — admits an interpretation as grounded in a single operational criterion: the existence of consistent local observers. The gauge-invariant norm $\Phi$ inherits this interpretation: it measures the irreducible obstruction as seen by any observer who satisfies these consistency requirements. This perspective is developed in [14, Section 1.2] and parallels the combinatorial and functional-analytic versions established in [16, Section 3.3] and [15, Section 6.4].

### 11.4 Limitations and Future Work

The inflation analysis in Section 6 is restricted to cycle graphs $C_n$ with a single-edge localized defect. A complete characterization of the inflation factor for general graphs remains open. The TV denoising analogy in Section 8 is structural rather than rigorous; formalizing the correspondence and establishing convergence rates is left to future work. The restriction to graphs without 2-cells means the coexact term vanishes. Extending the analysis to simplicial 2-complexes, where the full tripartite Hodge decomposition $\delta = d\alpha + \delta\beta + h$ is active, is a natural next step.

### 11.5 Closing Remark

The results of this paper do not introduce new optimization problems or algorithms; rather, they identify the structural role of an existing $\ell^1$ minimization within a cohomological framework. When combined with the preceding classification results [14–16], this shows that the measurement, decomposition, and optimization of defect fields are not independent choices, but are constrained by a single underlying geometric requirement. The $\ell^1$ quotient norm $\Phi$ thus serves as the consistent endpoint of this structure, linking local defect aggregation, global topology, and convex optimization in a common framework.

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## Acknowledgments

No external funding. No conflicts of interest.


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## References


[1] B. Eckmann, "Harmonische Funktionen und Randwertaufgaben in einem Komplex," *Commentarii Mathematici Helvetici* **17** (1944/45), 240–255.

[2] A. Chambolle and T. Pock, "A first-order primal-dual algorithm for convex problems with applications to imaging," *J. Math. Imaging Vision* **40** (2011), 120–145.

[3] L. R. Ford and D. R. Fulkerson, *Flows in Networks*, Princeton University Press, 1962.

[4] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.

[5] E. J. Cand'es and T. Tao, "Decoding by linear programming," *IEEE Trans. Inform. Theory* **51** (2005), 4203–4215.

[6] J. D. Horton, "A polynomial-time algorithm to find the shortest cycle basis of a graph," *SIAM J. Comput.* **16** (1987), 358–366.

[7] D. Horak and J. Jost, "Spectra of combinatorial Laplace operators on simplicial complexes," *Adv. Math.* **244** (2013), 303–336.

[8] L.-H. Lim, "Hodge Laplacians on Graphs," *SIAM Review* **62** (2020), 685–715.

[9] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970.

[10] R. Grady and A. Polimeni, *Discrete Calculus*, Springer, 2010.

[11] M. Gromov, "Volume and bounded cohomology," *Publications Math'ematiques de l'IH'ES* **56** (1982), 5–99.

[12] T. K. Dey, A. N. Hirani, and B. Krishnamoorthy, "Optimal homologous cycles, total unimodularity, and linear programming," *SIAM J. Comput.* **40** (2011), 1026–1044.

[13] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Physica D: Nonlinear Phenomena* **60** (1992), 259–268.

[14] J. H. Carroll, "Single Obstruction Theory: Retraction Nonlinearity, $\ell^1$ Rigidity, and Density Scaling," Zenodo, 2026. https://doi.org/10.5281/zenodo.19151803

[15] J. H. Carroll, "The Free $\ell^1$ Seminorm on Banach Presheaf Coboundaries," Zenodo, 2026. https://doi.org/10.5281/zenodo.19152115

[16] J. H. Carroll, "Coordinate-Wise Additivity and the $\ell^1$ Norm on Finite Graph Cochains," Zenodo, 2026. https://doi.org/10.5281/zenodo.18945342

[17] S. Kakutani, "Concrete representation of abstract $(L)$-spaces and the mean ergodic theorem," *Annals of Mathematics* **42** (1941), 523–537.